Thermodynamics of Darwinian selection in molecular replicators

Artemy Kolchinsky

SUPPLEMENTARY MATERIAL

Appendix A: Degradation reactions

Here we show that our thermodynamic bounds can be generalized to account for degradation reactions.

Let us consider the case where the replicator X undergoes degradation. In the absence of degradation, the replicator's concentration changes as

$$\dot{x} = \mathcal{J} - \phi x = r(\rho)x - r^{-}(\rho)x^2 - \phi x, \tag{S1}$$

where $\mathcal{J} = r(\rho)x - r^{-}(\rho)x^2$ is the net flux across the autocatalytic reaction. Now suppose that the replicator undergoes a degradation reaction,

$$X + \sum_i \alpha_i' A_i \to \sum_i \beta_i' A_i\,, \tag{S2}$$

where α_i' and β_i' are stoichiometric coefficients. We assume that the degradation is irreversible and that the flux across the degradation reaction is first order with rate constant $\eta \geq 0$, so ηx. As usual, the rate constant η may depend on the concentrations of substrates/side products $\vec{a}$, though we leave this implicit.

In the presence of degradation, the replicator's concentration changes as

$$\dot{x} = r(\rho)x - r^{-}(\rho)x^2 - \eta x - \phi x. \tag{S3}$$

We assume that $\eta < r(\rho)$, since otherwise positive growth is impossible. We then define a modified forward rate constant,

$$\tilde{r}(\rho) := r(\rho) - \eta \geq 0, \tag{S4}$$

so that Eq. (S3) becomes

$$\dot{x} = \tilde{r}(\rho)x - r^{-}(\rho)x^2 - \phi x, \tag{S5}$$

thereby recovering the form of Eq. (S1). The flux-force relation, Eq. (3.6) in the main text, becomes an inequality,

$$\sigma = \ln \frac{r(0)}{xr^{-}(0)} \geq \ln \frac{\tilde{r}(0)}{xr^{-}(0)}, \tag{S6}$$

since $\tilde{r}(0) \leq r(0)$. We can then define fitness $\tilde{f}$ in the same way as Eq. (4.1), except using $\tilde{r}(\rho)$ rather than $r(\rho)$. The derivation of our bounds in Section 5 in the main text proceeds in the same manner as before using these new definitions. Note that the derivative relations in Eq. (3.16) still apply to $\tilde{r}(\rho)$, since η does not depend on ρ.

Let us now consider the case of a non-elementary replicator X where some intermediate species Y_k also undergo irreversible degradation with rate constants $\pi_k \in [0, \nu_k)$. To account for this, the matrix $\bar{M}$ in the linear system (3.12) turns into

$$\bar{M}_{ij} := M_{ij} + \delta_{ij}\pi_i. \tag{S7}$$

Matrix calculus shows that

$$\frac{\partial \bar{M}^{-1}_{jk}}{\partial \bar{M}_{ii}} = -\bar{M}^{-1}_{ji}\bar{M}^{-1}_{ik}. \tag{S8}$$

For any set of degradation rates π_i, $\bar{M}$ is an M-matrix, so it is invertible and all entries of its inverse are nonnegative [2, 4]. Therefore, Eq. (S8) implies that every entry of $\bar{M}^{-1}$ decreases with increasing π_k. This means that the effective rate constants $\bar{r}(\rho)$ and $\bar{r}^{-}(\rho)$, defined as in Eq. (3.14) using $\bar{M}$, obey

$$\bar{r}(\rho) \leq r(\rho) \qquad \bar{r}^{-}(\rho) \geq r^{-}(\rho).$$

To account for the degradation of the replicator itself with rate η, we can further modify the effective forward rate constant as

$$\tilde{\bar{r}}(\rho) := \bar{r}(\rho) - \eta,$$

similarly to Eq. (S4). As above, we have a flux-force inequality,

$$\sigma = \ln \frac{r(0)}{xr^{-}(0)} \geq \ln \frac{\bar{r}(\rho)}{x\bar{r}^{-}(\rho)} \geq \ln \frac{\tilde{\bar{r}}(\rho)}{x\bar{r}^{-}(\rho)}.$$

The derivation of our bounds then proceeds as before, after defining fitness $\tilde{\bar{f}}$ using the forward rate constant $\tilde{\bar{r}}(\rho)$. Note that the derivative relations in Eq. (3.16) still apply to $\tilde{\bar{r}}(\rho)$, since $\bar{M}$ is an M-matrix and η does not depend on ρ.

In general, our thermodynamic bounds will tend to be less tight in the presence of degradation, since the flux-force equality (3.6) turns into an inequality. At the same time, the affinity of replication may increase due to degradation, for instance because the steady-state concentration of replicator decreases. Some of our bounds may also be larger, for instance the right side of the steady-state bound (5.2) will be larger, since fitness will decrease due to degradation.

Appendix B: Non-elementary replicators

Here we derive closed-form expressions of the effective rate constants in the $\rho = 0$ regime, as appear in Eq. (3.17). As discussed near the end of Section 3 (b) in the main text, this regime applies when the replication rate ρ is much slower than the rate of internal reactions.

Plugging $\rho = 0$ into Eq. (3.11) implies that $J_k = J_{k+1}$, so all intermediate fluxes are equal:

$$J_k = J \qquad \text{for } k \in 1..m. \tag{S9}$$

They also all equal to the rate of production of the replicator, $\mathcal{J} = 2J_m - J_1 = J$. We can then rearrange Eq. (3.7) to write the steady-state concentrations of Y_k as

$$y_k = (\nu_k/\nu_k^-)y_{k-1} - \mathcal{J}/\nu_k^- \qquad k \in 1..m,$$

where we use the convention $y_0 = x$, $y_m = x^2$. This is a first-order linear recurrence relation for y_k. Using an existing result [Thm. 7.1, 1], this recurrence can be solved for $y_m = x^2$ starting from the initial condition $y_0 = x$ to give

$$x^2 = \left(x - \mathcal{J} \sum_{k=1}^{m} \frac{\prod_{l=1}^{k-1} \nu_l^-}{\prod_{l=1}^{k} \nu_l} \right) \prod_{k=1}^{m} \frac{\nu_k}{\nu_k^-} \tag{S10}$$

where we use the convention $\prod_{l=1}^{k-1} \nu_l^- = 1$ for $k = 1$. Plugging in $r(0)$ and $r^-(0)$ from Eq. (3.17) gives

$$x^2 = (x - \mathcal{J}/r(0)) \frac{r(0)}{r^-(0)}.$$

This can be rearranged as $\mathcal{J} = r(0)x - r^-(0)x^2$, which recovers the mass-action-like form of Eq. (3.5).

Appendix C: Chemostat model

1. Steady state

Here we analyze the steady-state behavior of the dynamical system described by Eq. (7.2). To begin, let $\omega := a + \sum_i x_i$ indicate the total concentration of substrate and replicators at time t. The first line of Eq. (7.2) means that $k_i x_i (a - e^{\Delta G_i^\circ/RT} x_i) = \dot{x}_i + \phi x_i$. Plugging this into the second line of Eq. (7.2) and rearranging gives

$$\dot{\omega} = \phi(\gamma - \omega).$$

Thus, ω converges exponentially fast to the steady-state value $\omega^* = \gamma$.

We consider the long-term dynamics of the system restricted to the invariant subspace $\omega = \gamma$. Within this subspace, we can rewrite the first line of Eq. (7.2) as

$$\dot{x}_i = k_i x_i \big(\gamma - \sum_j x_j - e^{\Delta G_i^\circ/RT} x_i\big) - \phi x_i. \tag{S11}$$

Using an appropriate Lyapunov function, Schuster and Sigmund demonstrated that the dynamics in Eq. (S11) converge to the steady state in Eq. (7.3) for any strictly positive initial condition $(x_1(0), \dots, x_N(0)) \in \mathbb{R}_+^N$ [3]. A similar global convergence result can also be derived from the theory of Lotka-Volterra dynamics [5]. Specifically, Eq. (S11) can be written as a competitive Lotka-Volterra system,

$$\dot{x} = b_i x_i + \sum_j R_{ij} x_i x_j, \tag{S12}$$

where $b_i := k_i\gamma - \phi$ and $R_{ij} = -k_i(1 + \delta_{ij} e^{\Delta G_i^\circ/RT})$. The matrix R can be expressed as $R = -K(\vec{1}\vec{1}^T + D)$, where $K_{ij} = \delta_{ij} k_i$ and $D = \delta_{ij} e^{\Delta G_i^\circ/RT}$ are diagonal matrices and $\vec{1}$ is a vector of all 1s. Note that $\vec{1}\vec{1}^T + D$ is positive definite, since $\vec{1}\vec{1}^T$ is positive semidefinite and D is positive definite. It is known that for this type of Lotka-Volterra system, any strictly positive initial condition converges to a unique globally attracting fixed point [5], which is the steady state specified by Eq. (7.3).

The steady state in Eq. (7.3) is expressed as a set of coupled equations, which can be solved in the following manner. Assume without loss of generality that the rate constants k_i are arranged in decreasing order, $k_1 \geq k_2 \geq \cdots \geq k_N$. Given Eq. (7.3), it must then be that $x_i^* = 0$ implies $x_j^* = 0$ whenever $j > i$. Suppose for the moment that the top $i \in \{0..N\}$ replicators have non-zero steady-state concentrations,

$$x_j^* = \begin{cases} e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j) & j \leq i \\ 0 & j > i \end{cases} \tag{S13}$$

Eq. (7.3) gives $a^* = \gamma - \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j)$, so

$$a^* = \frac{\gamma + \phi \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}}. \tag{S14}$$

Eqs. (S13) and (S14) are the solution to Eq. (7.3) if for all $j \in \{0..N\}$,

$$x_j^* = \max\{0, e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j)\}. \tag{S15}$$

Given Eq. (S13), Eq. (S15) is satisfied once

$$a^* - \phi/k_i \geq 0 \geq a^* - \phi/k_{i+1}, \tag{S16}$$

Therefore, to solve Eq. (7.3), it suffices to evaluate Eqs. (S14) and (S13) for $i = 0, 1, 2, \dots$, stopping once the bounds (S16) are satisfied.

2. Derivation of inequality (7.4) (condition for extinction)

Given Eq. (7.3), replicator X_i is extinct once

$$a^* \leq \phi/k_i. \tag{S17}$$

If this inequality holds, then any lower fitness replicator X_j ($k_j \leq k_i$) must also be extinct, since then $a \leq \phi/k_j$. Now, combine the equations in Eq. (7.3) to write

$$\begin{aligned} a^* &= \gamma - \sum_{j:x_j>0} e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j) \\ &\leq \gamma - \sum_{j:k_j>k_i} e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j), \end{aligned} \tag{S18}$$

where the inequality in the second line reflects that it may be that $a^* \leq \phi/k_j$ even for higher fitness replicators ($k_j > k_i$). Rearranging inequality (S18) gives

$$a \leq \frac{\gamma + \phi \sum_{j:k_j \geq k_i} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j:k_j \geq k_i} e^{-\Delta G_j^\circ/RT}}. \tag{S19}$$

Given inequality (S19), the bound (S17) must be satisfied when

$$\frac{\gamma + \phi \sum_{j:k_j \geq k_i} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j:k_j \geq k_i} e^{-\Delta G_j^\circ/RT}} \leq \phi/k_i.$$

Rearranging gives the inequality (7.4).

3. Nonequilibrium phase transitions at extinctions

We show that the entropy production rate, Eq. (7.6), is continuous but not differentiable at extinction events, meaning that extinctions are second-order nonequilibrium phase transitions.

Suppose that the rate constants are strictly ordered as

$$k_1 > k_2 > \cdots > k_N. \tag{S20}$$

From Eq. (7.4), the critical dilution rate for replicator X_i is

$$\hat{\phi}_{(i)} = \frac{\gamma}{k_i^{-1} + \sum_{j=1}^{i-1} e^{-\Delta G_j^\circ/RT}(k_i^{-1} - k_j^{-1})} \tag{S21}$$

$$= \frac{\gamma}{k_i^{-1} + \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}(k_i^{-1} - k_j^{-1})} \tag{S22}$$

where in the second line we used $k_i^{-1} - k_j^{-1} = 0$ for $j = i$. We treat ϕ as the control parameter, while holding γ fixed.

We show that the steady-state substrate concentration a^* is continuous as a function of the dilution rate at $\hat{\phi}_{(i)}$. When $\phi < \hat{\phi}_{(i)}$, replicators $X_1, \ldots, X_i$ are not extinct. We consider the limit from below,

$$\lim_{\phi \nearrow \hat{\phi}_{(i)}} a^* = \frac{\gamma + \hat{\phi}_{(i)} \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}} = \hat{\phi}_{(i)} k_i^{-1}.$$

Here we first used Eq. (S14) and then plugged in Eq. (S21) and simplified using a bit of tedious algebra. When $\phi > \hat{\phi}_{(i)}$, replicators $X_1, \ldots, X_{i-1}$ are not extinct, but replicator X_i is extinct. We consider the limit from above,

$$\lim_{\phi \searrow \hat{\phi}_{(i)}} a^* = \frac{\gamma + \phi \sum_{j=1}^{i-1} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j=1}^{i-1} e^{-\Delta G_j^\circ/RT}} = \hat{\phi}_{(i)} k_i^{-1},$$

where we first used Eq. (S14) and then plugged in Eq. (S22) and simplified. The two limits match, so a^* is continuous at $\hat{\phi}_{(i)}$. This also implies that steady-state replicator concentrations $x_j^* = \max\{0, e^{-\Delta G_j^\circ/RT}(a^* - \phi/k_j)\}$ from Eq. (7.3) are continuous at $\hat{\phi}_{(i)}$.

Next, consider the entropy production rate at the critical point,

$$\lim_{\phi \to \hat{\phi}_{(i)}} \dot{\Sigma} =$$

$$\lim_{\phi \to \hat{\phi}_{(i)}} \phi \left[\sum_{j=1}^{i-1} x_j^* \Big(\ln \frac{a^*}{x_j^*} - \frac{\Delta G_j^\circ}{RT} \Big) + x_i^* \Big(\ln a^* - \frac{\Delta G_i^\circ}{RT} \Big) \right],$$

where we used $\lim_{\phi \to \hat{\phi}_{(i)}} x_i^* \ln x_i^* = \lim_{\alpha \to 0} \alpha \ln \alpha = 0$. Given Eq. (S20), $x_j^* > 0$ for $j \in \{1..i-1\}$, so all the terms on the right side are finite and continuous at $\phi = \hat{\phi}_{(i)}$. Thus, $\dot{\Sigma}$ is a continuous function of ϕ at $\hat{\phi}_{(i)}$.

Next, we show that $\dot{\Sigma}$ is not differentiable with respect to ϕ at $\hat{\phi}_{(i)}$ because it has an infinite left derivative at this point (it is not hard to show that the right derivative is finite). The left derivative is

$$\partial_\phi^- \dot{\Sigma} = \sum_{j=1}^{i} \left[\partial_\phi^- x_j^* \Big(\ln a^* - \ln x_j^* - 1 - \frac{\Delta G_j^\circ}{RT} \Big) + \frac{x_j^*}{a^*} \partial_\phi^- a^* \right].$$

All terms in this expression are finite except for $-(\partial_\phi^- x_i^*) \ln x_i^*$. Plugging in Eq. (7.3) gives

$$\begin{aligned}
\partial_\phi^- x_i^* &= e^{-\Delta G_i^\circ/RT}(\partial_\phi^- a^* - k_i^{-1}) \\
&= e^{-\Delta G_i^\circ/RT} \left[\frac{\sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT} k_j^{-1}}{1 + \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}} - k_i^{-1} \right] \\
&= e^{-\Delta G_i^\circ/RT} \left[\frac{\sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}(k_j^{-1} - k_i^{-1}) - k_i^{-1}}{1 + \sum_{j=1}^{i} e^{-\Delta G_j^\circ/RT}} \right] \\
&< 0,
\end{aligned}$$

where in the second line we used Eq. (S14), and in the last line we used that $k_j > k_i$ and $k_i > 0$. Hence, $\partial_\phi^- x_i^*$ is a strictly negative constant, meaning that $\phi \nearrow \hat{\phi}_{(i)}$, and $x_i^* \to 0$, $-(\partial_\phi^- x_i^*) \ln x_i^* \to -\infty$. This shows that the left derivative of $\dot{\Sigma}$ is negative infinite at $\phi = \hat{\phi}_{(i)}$.

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